Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	Biosensor Nanovesicles		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	E.Almeida@nasa.gov	**Fax:**	FY 650-604-3159
PI Organization Type:	NASA CENTER	**Phone:**	650-604-1772
Organization Name:	NASA Ames Research Center		
PI Address 1:	MS 236-7		
PI Address 2:	Building 236 Room 217		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	07/01/2001	**End Date:**	06/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Fung, Paul	**Contact Phone:**	650-604-1276
Contact Email:	pfung@mail.arc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Almeida, Eduardo		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

We propose to develop technology for a novel biosensor consisting of lipid nanovesicles containing engineered reporter enzymes, for the in situ detection of cell surface antigens and other biomolecular markers. The detection system that we propose to develop will incorporate new technology for generating signals by bringing inactive fragments of an enzyme together inside a lipid vesicle, thereby yielding enzyme activation and signal amplification. Although there are a variety of vesicle-based, biomolecular sensors currently under development, no other technology takes advantage of the signal amplification that these types of engineered enzymes can provide. The use of well-characterized integrins as the signal transduction machinery in this system will provide a readily accessible means for demonstrating proof of concept, and will pave the way for further testing and development of clinical applications. Biosensor Nanovesicles targeted against cancer-specific antigens, for example, could be used as injectable tracers, that could be detected by fluorescence-based whole body imaging techniques or organ-specific imaging techniques (for detection and diagnosis of lymphoma, breast cancer, lung cancer, and colon cancer). Localized delivery of Biosensor Nanovesicles could also be accomplished by a

Task Description:	catheter-based delivery system, in order to provide improved anatomical definition of the extent of tumor involvement within a single organ system (such as brain tumors, ovarian cancer, uterine cancer, and prostate cancer). Biosensor Nanovesicles targeted against bacterial antigens or viral antigens could also provide a powerful tool for diagnosis of a variety of infectious diseases and would be well suited for detection of microbial pathogens that are difficult to isolate by conventional culture techniques. With the use of cryoprotectants, lipid vesicles can be freeze-dried and stored for years, with full preservation of biological activity. Thus, Biosensor Nanovesicles, may become useful biomedical tools for long-term space travel, in the form of microfluidic diagnostic testing platforms. Objectives: Aim 1) To generate DNA constructs encoding chimeric proteins with integrin and b-lactamase complementary enzyme fragments. Recombinant DNA technology will be used to produce chimeric proteins consisting of a4 or a5 integrins linked to artificial complementary fragments of the enzyme, b-lactamase. The chimeric proteins are designed to produce a fluorescent signal once enzyme activity is reconstituted via protein fragment complementation. Aim 2) To express chimeric b-lactamase enzyme fragments in mammalian cells. Cells that are defective in their expression of endogenous integrins will be infected with adenoviruses engineered to express the chimeric proteins at very high levels (Aim1) Aim 3) To induce chimeric b-lactamase enzyme fragment complementation and activation in cells using peptide-mediated integrin clustering. Cells that overproduce the integrin-enzyme chimeras (Aim 2) will be treated with peptides that bind and cluster integrins, thus producing b-lactamase fragment complementation and reconstitution of enzyme activity, to be detected by fluorogenic substrates. The peptides to be used include poly-LDV (which binds a4 integrin), and poly-RGD (which binds a5 integrin). Together, this provides the signal detection and reporting mechanism for cell-based systems. Aim 4) To generate lipid vesicles that contain integrin/enzyme chimeras localized to the membrane. In this aim, the b-lactamase enzyme fragment complementation and activation in cells will be expressed and purified from mammalian cells, then transferred to artificial lipid nanovesicles. Aim 5) To induce chimeric b-lactamase enzyme fragment complementation and activation in lipid nanovesicles using peptide-mediated integrin clustering. In this aim, cell based integrin clustering based protein fragment complementation and activation of the b-lactamase reporter system will be transferred from mammalian cells to lipid nanovesicles. The Biosensor Nanovesicles that we will develop will integrate molecular recognition with enzymatic signal amplification to create a molecular detection system with high specificity and high sensitivity. The system is designed to be a versatile platform, suitable for a broad range of molecular detection needs. An important application of Biosensor Nanovesicles will be cancer detection. Biosensor Nanovesicles targeted against cancer-specific antigens, for example, could be used as injectible tracers that could be detected by fluorescence-based whole body imaging techniques or organ-specific imaging techniques (for detection and diagnosis of lymphoma, breast cancer, lung cancer, and colon cancer and other solid tumors). Localized delivery of Biosensor Nanovesicles could also be accomplished by a catheter-based delivery system, in order to provide improved anatomical definition of the extent of tumor involvement within a single organ system (such as brain tumors, ovarian cancer, uterine cancer and, prostate cancer). The technology we propose could be further developed to combine molecular sensing with a vesicle-based drug targeting and delivery system. Triggered activation of an enzyme that disrupts lipid vesicle integrity, such as a lipase, could form the basis for release of vesicle contents at specific sites within the body. Localized release of both a drug and an imaging agent could provide a way to deliver a localized therapeutic agent with a built-in system for verifying delivery or monitoring response to therapy. Biomedical applications of this technology extend beyond the field of oncology, and include possible use as a platform for in vitro diagnostic testing. Biosensor Nanovesicles have the potential to be assembled into microfluidic arrays, to create devices with the capability of simultaneously sensing multiple cell surface antigens or even bacterial or viral markers. Gene sequences could be targeted as well, since microfluidic devices have the capability of breaking open cells to release their DNA content. Labor-intensive tests that are currently performed as part of multi-step analytical procedures (such as flow cytometry, enzyme-linked immunoabsorbant assays (ELISA), and molecular diagnosis by PCR) could be incorporated into a microfluidic platform of this type, with the benefits of miniaturization, rapid turn-around and low-cost. This type of application would fulfill NASA and the National Cancer Institute's shared vision of technology that could revolutionize the speed and effectiveness of basic health care on Earth, as well as provide the medical technology needed for long-duration human space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress provided this reporting period.
Bibliography Type:	Description: (Last Updated: 12/08/2025)